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Dear IFAE,

My interest in theoretical physics has been driven by a desire to understand the fundamental structure of nature, which has naturally led me toward quantum field theory and particle physics. I am particularly fascinated by the role of symmetries in shaping physical laws, including how phenomena such as spontaneous symmetry breaking and anomalies determine particle behaviour. I wish to deepen my understanding of these principles and ultimately piece the clues together to answer: what symmetries govern physics beyond the Standard Model?

My research experience spans fundamental particle physics, cosmology, and computational methods. Currently, I am working on my master's thesis, which focuses on gaining a deeper understanding of fundamental interactions and of beyond-the-Standard-Model physics. In my research, I explore grand unified theories (GUTs). Rather than proposing a specific gauge group, I study a general landscape of high-energy GUTs and identify preferred models by minimizing their free energy. In parallel, I am involved in a project on infinite heat order, a framework related to high-temperature behaviour and symmetry non-restoration in UV-complete models. Through this work with my supervisor, Francesco Sannino, I have developed a strong foundation in quantum field theory.

Besides my thesis, my research experience includes a project on anomalies in quantum field theory. During my bachelor's degree, I worked on density profiles of dark matter, both theoretically and using machine learning models. This interest culminated in a bachelor's thesis in cosmology, where I examined whether inhomogeneities in the structure of the universe could explain its accelerated expansion.

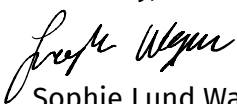
Across these projects, I have developed a strong foundation in theoretical physics and computational methods, which I am eager to apply to further study of fundamental questions.

For my PhD, I hope to contribute to research in theoretical particle physics. I yearn to use the framework of quantum field theory to study the interplay of flavour, Higgs dynamics, and beyond-Standard-Model physics, with the hope of uncovering the deeper principles governing fundamental interactions. I am also motivated by open questions at the interface of particle physics and cosmology, and by the possibility of glimpsing information about the early universe through effective field theory methods. Ultimately, I aim to help clarify the underlying principles that determine the structure of our universe.

I am strongly motivated to apply to the IFAE due to its collaborative environment and its strong links between theory and experiment, which provide an ideal setting for PhD training. I am particularly attracted to the Theory Group at IFAE due to its strong activity in physics beyond the Standard Model and cosmology, especially in approaches based on effective field theories and their applications to early-universe physics and phenomenology. I would be especially interested in learning from and contributing to research directions pursued by members such as Prof. A. Pomarol and Prof. J. Matias.

Thank you for considering my application. I would be thrilled to contribute to the research activities of the Theory Group at the IFAE.

Sincerely,



Sophie Lund Wagner (ID-number: 211676172)